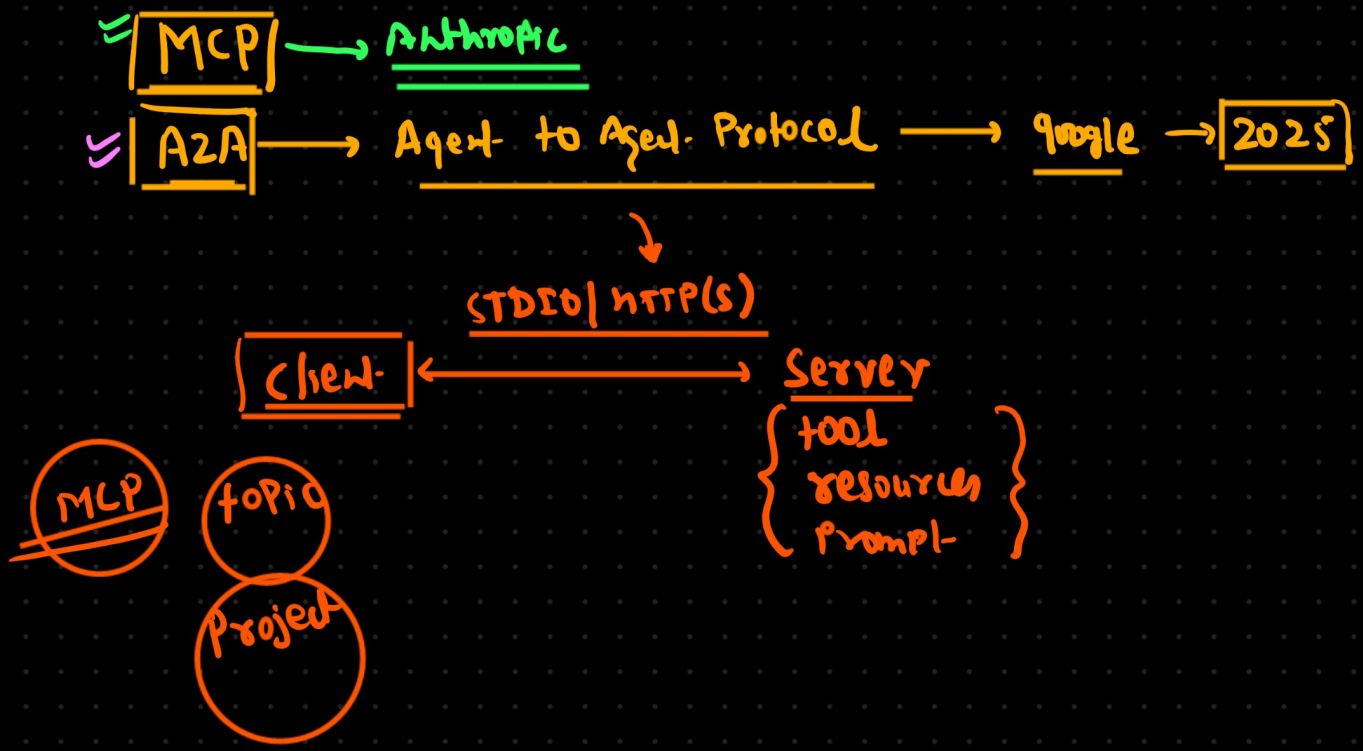
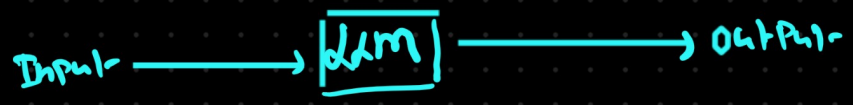

MCP
Anthropic
A2A
Agent to Agent Protocol
google
2025
STDIO | HTTP(S)
Client
Server
tool
resources
prompt
MCP
topic
project

AI System



Not having a recent knowledge

Extrenal knowledge → Context

Standardized

client/server

MCP

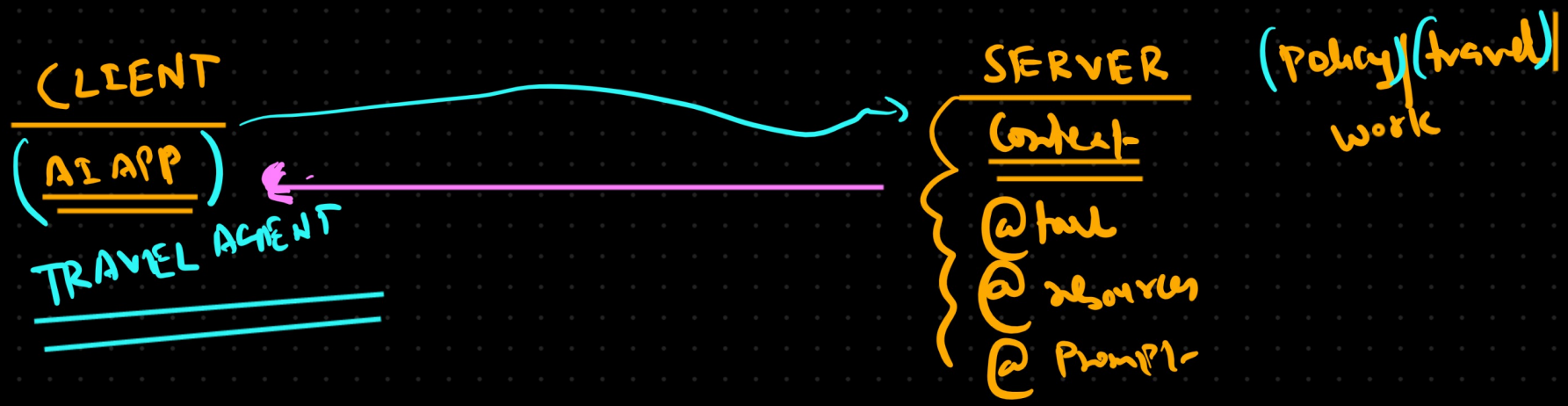
Model | Context | Protocol

Retrieval

@tool

@resources

Outside knowledge


CLIENT
AI APP
TRAVEL AGENT
SERVER
Context
@tool
@resources
@prompts
(policy)(travel)
work

A2A ⟶ Agent to Agent Protocol

Agent A

*

Agent B

Custom Integration

Agent-A ⟷ Agent B

Agent A ⟷ Agent C

Custom Integration

Phase 1

LLM ⟶ Answer

Phase 2

LLM ⟶ Tools ⟶ API / DB / Rag

Phase 3



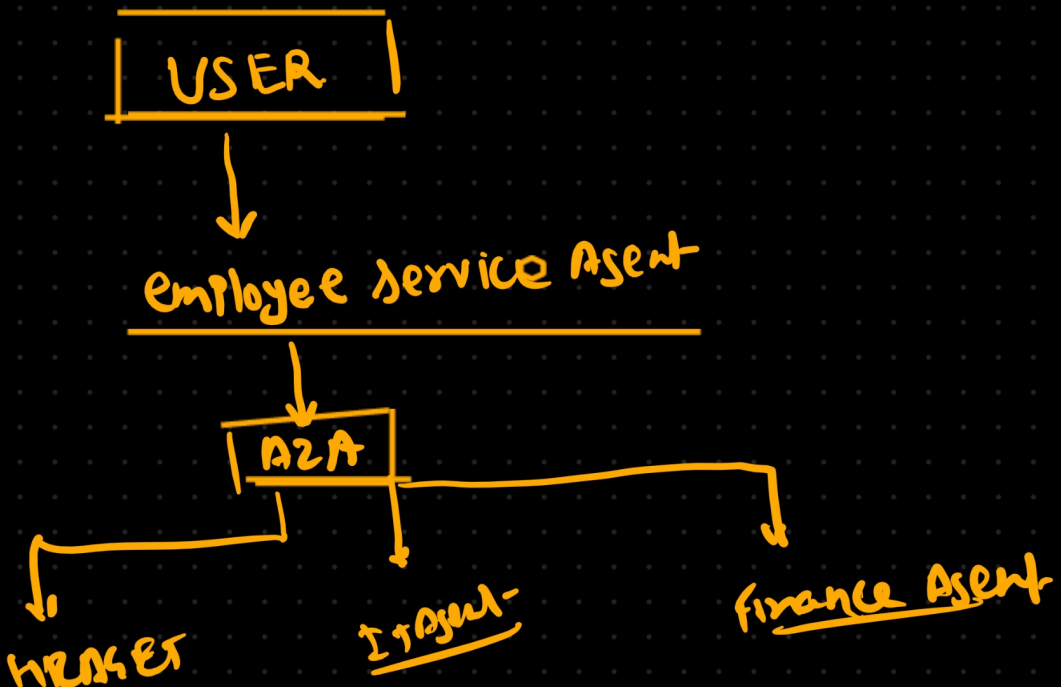
Phase 4

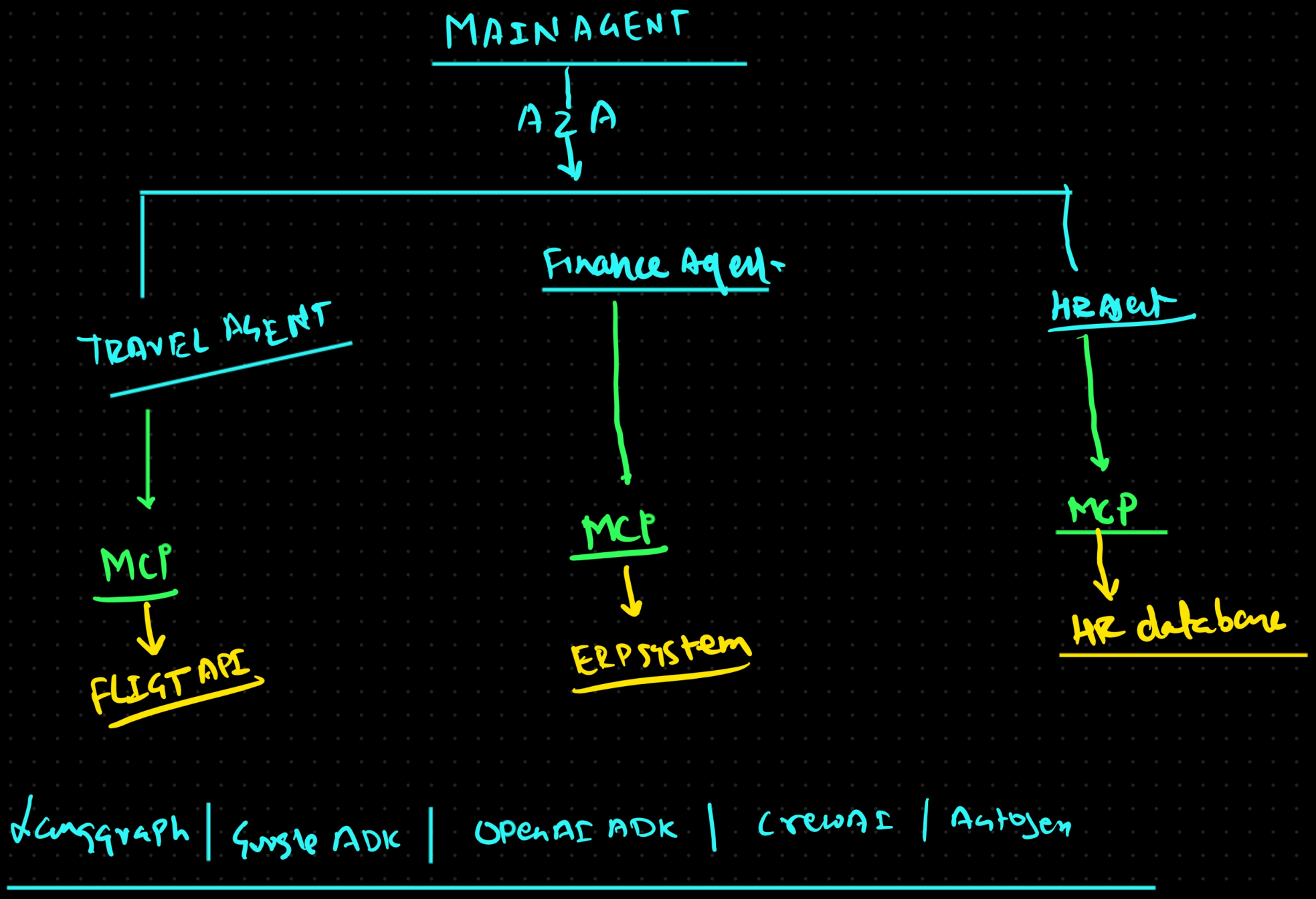
TRAVEL Agent-

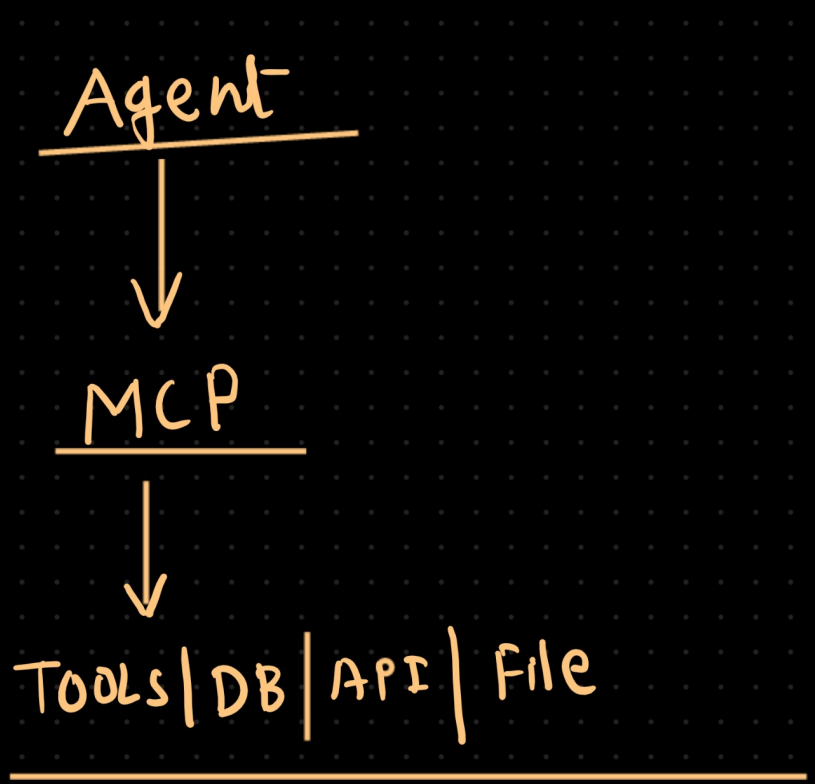
Finance Agent-

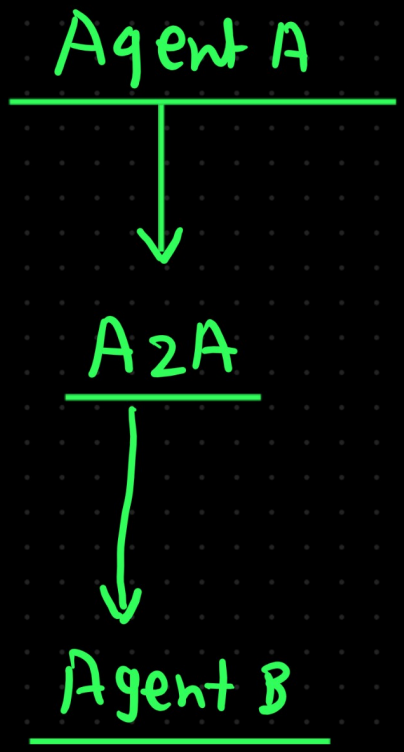
HR Agent-

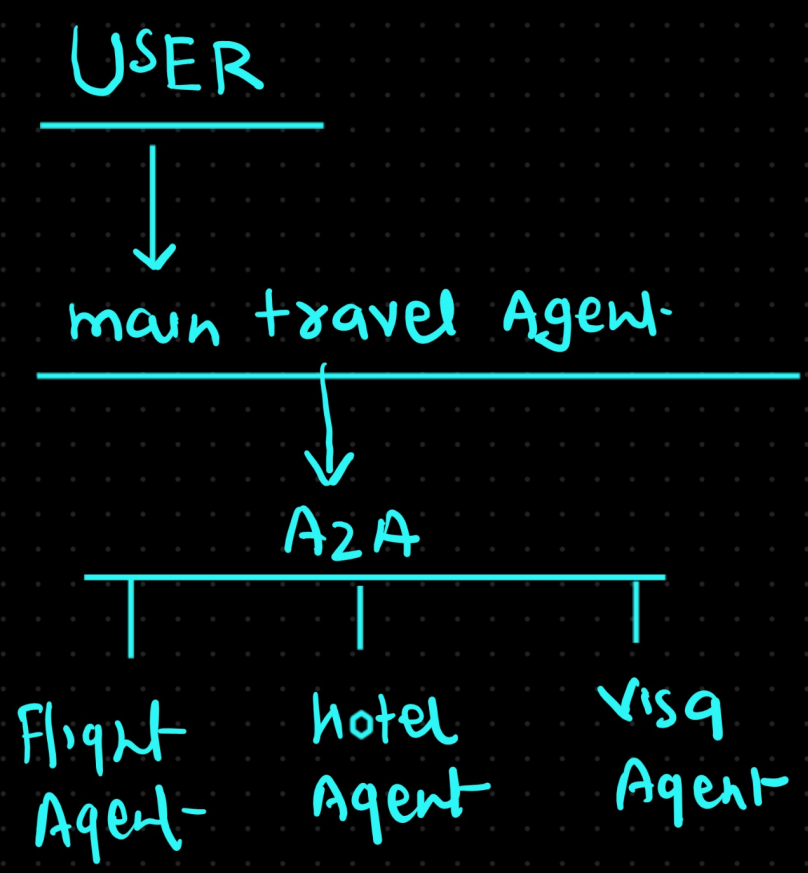


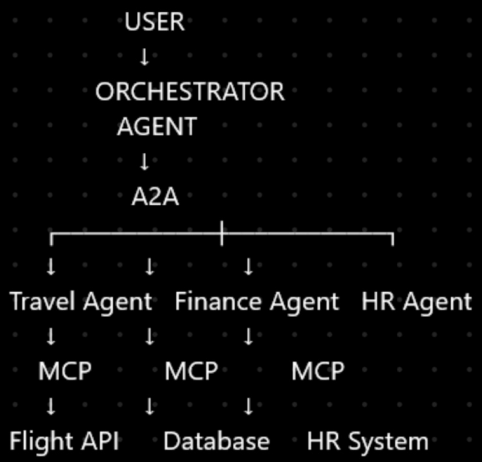



MAIN AGENT
A2A
TRAVEL AGENT
Finance Agent
HR Agent
MCP
MCP
MCP
FLIGT API
ERP system
HR database
Langgraph | Google ADK | OpenAI ADK | CrewAI | Agno

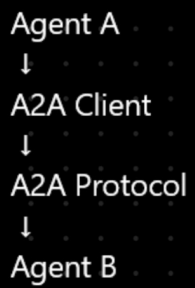

Agent
MCP
TOOLS | DB | API | File


Agent A
A2A
Agent B


USER
main travel Agent
A2A
Flight Agent
hotel Agent
Visa Agent



A2A connects agents to agents. MCP connects those agents to tools and data



Travel Orchestrator Agent

↓

A2A Protocol

↓

JSON-RPC

↓

HTTP

↓

Flight Agent

JSON-RPC
gRPC
HTTP+JSON

Travel Orchestrator

↓

GET Flight Agent Card

↓

discovers:
Search Flights skill

↓

A2A message

↓

Flight Agent

↓

Flight Response

Travel Orchestrator

↓

GET Hotel Agent Card

↓

discovers:
Find Hotels skill

↓

A2A message

↓

Hotel Agent

↓

Hotel Response

2024-Anthropic

MCP

(Context which we Provide to model)

tool/resource/prompt-

Client- Server

2025-google

A2A

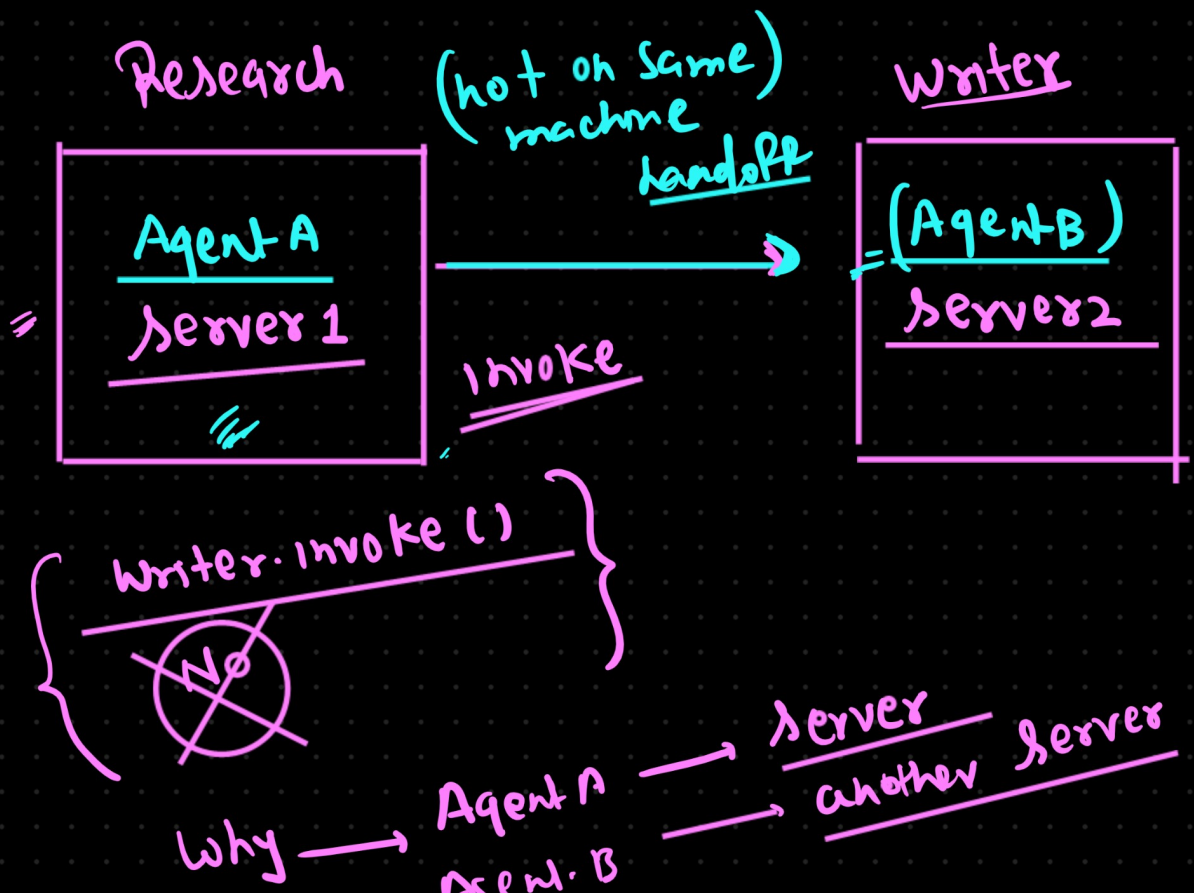
Practical

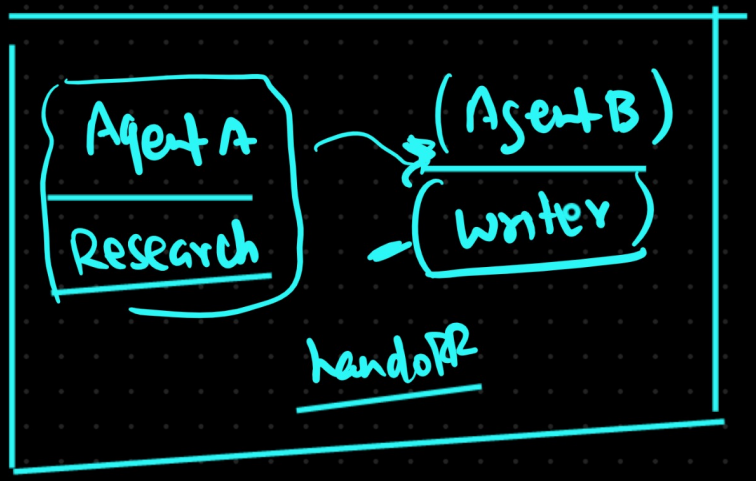
{agent 2 agent
multiagent-system}

Communication

standardize

{Client- Server}


Research
(not on same machine)
handoff
Writer
Agent A
Server 1
Agent B
Server 2
invoke
writer.invoke()
No
Why → Agent A → Server
Agent B → another Server



same application
same server ⟶ system | machine |


USER
"Plan Dubai Trip"
travel_orchestrator.py
Discover Flight Agent
Get Agent Card
Read Agent Skills
Create A2A Client
SendMessage
flight_agent.py
DefaultRequestHandler
RequestContext
AgentExecutor
Create Text Message
EventQueue
Flight Response
Discover Hotel Agent
Get Agent Card
Read Agent Skills
Create A2A Client
SendMessage
hotel_agent.py
DefaultRequestHandler
RequestContext
AgentExecutor
Create Text Message
EventQueue
Hotel Response
Display Both Results